

LONGREACH, Australia

WMO: 943460

Lat: 23.4398S

Lon: 144.2830E

Elev: 193

StdP: 99.03

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	5.8	-13.7	1.2	21.1	-11.0	1.5	20.2	9.4	20.7	8.8	19.9	2.5	140	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.3	40.8	20.4	39.5	20.2	38.3	20.2	24.9	30.6	24.3	30.4	23.7	30.2	4.7	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.1	26.4	23.1	18.2	26.1	22.2	17.3	26.1	77.3	30.2	74.6	30.3	72.3	29.9	35.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.9	8.1	DB	1.0	42.8	0.9	1.8	0.3	44.1	-0.2	45.2	-0.8	46.2	-1.5	47.5	(4)
(5)			WB	-1.2	26.9	1.3	2.0	-2.2	28.3	-2.9	29.5	-3.6	30.7	-4.5	32.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	29.9	29.0	27.8	24.2	19.6	16.3	15.9	17.5	22.0	25.8	28.1	29.9	(6)	
	DBStd	5.98	2.65	2.62	2.45	2.73	3.14	3.28	3.10	3.14	3.44	3.32	2.97	2.96	(7)	
	HDD10.0	1	0	0	0	0	0	1	1	0	0	0	0	0	(8)	
	HDD18.3	254	0	0	0	1	23	79	89	53	8	1	0	0	(9)	
	CDD10.0	5044	618	533	552	425	299	188	183	233	361	490	544	618	(10)	
	CDD18.3	2255	359	300	293	176	63	17	13	28	118	233	294	360	(11)	
	CDH23.3	29727	4998	3921	3632	1943	689	195	196	509	1578	3057	3929	5078	(12)	
(13)	CDH26.7	16436	3016	2266	1990	891	183	24	27	141	744	1704	2311	3140	(13)	
(14)	Wind	WSAvg	4.3	4.6	4.5	4.3	4.0	3.8	3.9	4.1	4.3	4.7	4.8	4.7	(14)	
(15) Precipitation	PrecAvg	425	74	76	51	32	22	19	17	11	13	20	31	56	(15)	
	PrecMax	875	420	244	413	314	109	171	112	72	131	72	183	193	(16)	
	PrecMin	106	2	6	0	0	0	0	0	0	0	0	0	4	(17)	
	PrecStd	170	77	61	65	58	30	30	28	16	23	21	42	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	41.5	39.5	35.8	32.3	28.9	29.4	33.0	37.5	40.3	41.3	42.3	(19)	
		MCWB	21.1	21.1	20.6	20.0	17.6	16.8	16.8	16.1	17.2	19.0	19.4	20.4	(20)	
	2%	DB	40.1	39.2	37.7	34.1	30.0	27.2	27.1	30.0	34.9	37.8	39.3	40.7	(21)	
		MCWB	21.0	21.2	20.4	18.9	17.1	16.4	15.2	15.6	16.4	18.4	19.2	20.1	(22)	
	5%	DB	38.6	37.6	36.1	32.7	28.7	25.8	25.5	28.0	32.9	36.1	37.8	39.2	(23)	
		MCWB	21.1	21.2	20.4	18.3	16.6	15.6	14.4	14.5	16.1	17.7	18.9	20.1	(24)	
	10%	DB	37.1	36.1	34.5	31.4	27.2	24.0	23.9	26.3	31.1	34.3	36.1	37.6	(25)	
		MCWB	21.2	21.2	20.2	18.0	16.0	14.6	13.5	13.7	15.8	17.3	18.8	20.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.0	25.7	24.9	23.0	20.7	19.0	19.7	18.6	21.3	22.1	23.9	24.7	(27)	
		MCDB	31.4	31.2	28.9	28.7	26.7	25.5	23.3	26.8	27.6	31.2	30.2	31.4	(28)	
	2%	WB	25.0	24.9	24.0	21.9	19.6	17.8	17.8	17.2	19.7	21.0	22.7	23.9	(29)	
		MCDB	30.6	30.7	29.2	28.4	25.7	24.0	23.6	25.8	26.9	30.1	29.4	30.9	(30)	
	5%	WB	24.2	24.2	23.2	20.9	18.5	16.9	16.2	16.0	18.4	20.2	21.9	23.2	(31)	
		MCDB	30.4	30.5	29.4	28.4	25.6	22.9	22.7	24.9	26.9	29.8	29.7	30.6	(32)	
	10%	WB	23.7	23.7	22.5	20.0	17.5	15.8	14.7	14.9	17.3	19.4	21.1	22.7	(33)	
		MCDB	30.4	30.2	29.5	27.5	25.1	22.0	22.2	24.3	27.6	29.5	29.8	30.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	12.4	13.0	13.9	14.5	14.4	15.5	16.7	16.5	15.6	14.2	13.3	(35)	
		MCDBR	14.8	14.6	14.7	15.0	15.3	15.4	16.7	18.4	18.4	17.3	16.1	15.6	(36)	
		MCWBR	5.2	4.6	5.3	6.1	6.9	7.3	8.3	9.0	7.9	7.2	6.1	6.0	(37)	
	5% WB	MCDBR	11.3	10.9	10.9	11.8	12.0	12.1	13.4	15.4	14.2	13.8	12.6	12.0	(38)	
		MCWBR	4.1	3.6	3.9	4.8	5.8	5.9	7.1	7.8	6.4	6.1	5.0	4.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.382	0.372	0.355	0.341	0.325	0.307	0.300	0.318	0.360	0.377	0.389	0.390	(40)	
	taud		2.474	2.500	2.528	2.522	2.507	2.562	2.560	2.484	2.380	2.360	2.375	2.410	(41)	
	Ebn at Noon		962	958	947	918	892	892	913	928	927	942	949	955	(42)	
	Edh at Noon		119	113	105	98	91	82	85	99	120	129	130	127	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.28	7.06	6.42	5.54	4.53	3.98	4.40	5.39	6.35	7.26	7.52	7.49	(44)	
	RadStd		0.51	0.43	0.44	0.41	0.22	0.26	0.26	0.25	0.47	0.47	0.44	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page